SHEEPCARE

INTELLIGENT DETECTION FOR SHEEP

User Manual · Version 1.5.0

WITH THE SUPPORT OF THE EUROPEAN UNION AND THE OPENAGRI PROJECT

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1. Introduction

SheepCare is a decision-support system for reproductive and production management in sheep. With it you can:

- **Detect estrus**: it analyses each animal's vaginal resistance measurements and classifies it as *Estrus* or *No estrus*.
- **Classify milk quality**: from the near-infrared spectrum of each sample, it reports whether the somatic cell count is of *high* or *medium* quality.
- **Check the weather and heat stress** for each farm, current and forecast for 5 days.
- **Generate PDF reports** for estrus and milk quality, per farm.
- **Record the results in the OpenAgri Farm Calendar**.

The local (Edge) version, described in this manual, stores all data on the device itself and the analysis works offline. Only the **cloud synchronisation** and the **weather lookup** need an Internet connection, and both are optional.

> **Note:** SheepCare is a decision-support tool. Results should always be interpreted together with the judgement of the responsible technician or veterinarian.

> **Language of the interface:** the application screens are in Spanish. In this manual, each button or screen is given with its Spanish label in quotation marks, followed by an English translation where needed.

2. System Requirements

Component	Minimum recommended
Operating system	Windows 10 (64-bit) or Windows 11
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	8 GB
Disk space	3 GB free (the application takes about 1.5 GB and the data grows with use)
Web browser	Google Chrome, Edge or Firefox (recent version)
Internet connection	Only for cloud synchronisation and the weather lookup

SheepCare is installed as a native Windows application. It does not require Docker or any other pre-installed software.

3. Installation

1. Download the installer **SheepCare-Setup-1.5.exe** (about 440 MB) and double-click it.
2. If Windows displays a security warning, click "More info" and then "Run anyway".
3. Follow the setup wizard steps. The installation does not require an Internet connection.
4. When finished, a SheepCare shortcut will appear on the desktop and in the Start menu.

Upgrading from a previous version

Close SheepCare, uninstall the previous version from Windows "Apps" and run the new installer. Your data (database, settings and credentials) is kept in the `%APPDATA%\SheepCare` folder and uninstalling does not delete it. The first time you open the new version, SheepCare adapts the database to the changes and, when it does so, first saves a backup copy (`.bak` files) in that same folder.

> **Changes from earlier versions:** milk quality is shown as *high* and *medium* (previously *good* and *poor*), and weather and heat stress, PDF reports and working with farms shared from the cloud have been added.

4. First Launch

Double-click the SheepCare icon on the desktop. A small progress window will appear showing the startup status:

- **Setting up the database** (*Configurando la base de datos*): first time only; it can take up to one minute.
- **Starting the database** and the **authentication service** (GateKeeper).
- **Starting the calendar module** (Farm Calendar).
- **Starting the weather service** and the **reporting service**.
- **Starting SheepCare** and, finally, **Ready! Opening SheepCare** (*¡Listo! Abriendo SheepCare*): the web browser opens automatically.

Subsequent launches are faster, since the databases are already initialised.

> **Application address:** SheepCare always opens at `http://localhost:8000` . You can return to this address from any browser while the application is running.

> **First-time setup only:** a small window will ask you to create the administrator account for GateKeeper, the shared login service used by the Calendar module (see section 5.9). Choose a username and a password of at least 8 characters; you will need them every time you open the Calendar. This step only appears once.

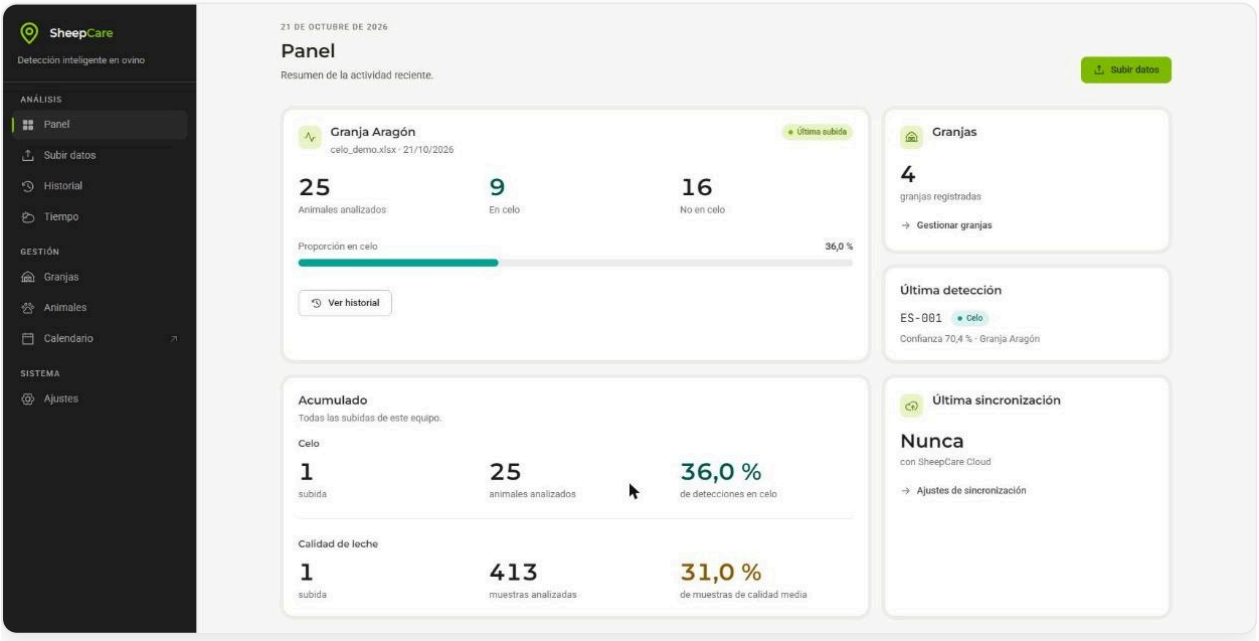
5. The Application Interface

The screen is divided into two areas: the left sidebar (navigation) and the content area on the right. The sidebar groups the screens into **Análisis** (Analysis: Panel, Subir datos, Historial, Tiempo), **Gestión** (Management: Granjas, Animales, Calendario) and **Sistema** (System: Ajustes). At the bottom you will find the installed version and the button to switch between light and dark mode.

5.1 Dashboard ("Panel")

The home screen. It summarises recent activity:

- **Last estrus upload**: farm, file, date, animals analysed, how many in estrus and how many not, and the proportion in estrus.
- **Farms** ("Granjas"): how many are registered, with a shortcut to manage them.
- **Last detection** ("Última detección"): the latest animal analysed, its result and its confidence.
- **Cumulative** ("Acumulado"): all uploads on this device, split into *Estrus* (uploads, animals analysed and percentage in estrus) and *Milk quality* (uploads, samples analysed and percentage of medium quality).
- **Last synchronisation** with the cloud, or "Nunca" (Never) if it is not configured.



Dashboard with an estrus upload (9 animals in estrus out of 25) and a milk quality upload (413 samples). Demonstration data.

5.2 Farms ("Granjas")

This section manages farms. Before uploading any data file, at least one farm must be registered.

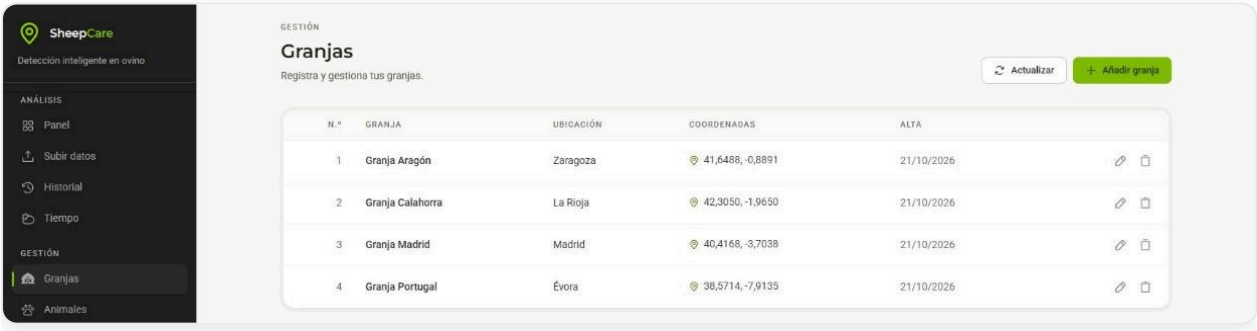
Create a farm

1. Click "Añadir granja" (Add farm).
2. Enter the farm name (required) and, optionally, its location.
3. Optionally, enter the **latitude and longitude** (for example, 42,8125 and -1,6458). With them you can look up the farm's weather in the "Tiempo" screen.
4. Click "Crear granja" (Create farm).



"Nueva granja" form with name, location and coordinates.

In the table, the icons in the actions column let you **edit** (pencil) or **delete** (bin) each farm. The table also shows the coordinates, or "Sin coordenadas" (No coordinates) if none were entered.



List of farms with location and coordinates.

> **Warning:** deleting a farm also deletes all its animals, uploads and results. This action cannot be undone.

Shared farms

If your cloud account has access to farms belonging to other people in your organisation, they appear in the table with the label **"Compartida contigo" (Shared with you)**. On a shared farm you

can view the data and upload new files, but you **cannot edit or delete it**, so the action icons are not shown. More details in section 7.

5.3 Upload Data: Estrus Detection

This is SheepCare's main function. It loads a vaginal resistance data file and returns the result for each animal. The "Subir datos" (Upload data) screen has two independent cards: *Detección de celo* (Estrus detection, left) and *Calidad de leche* (Milk quality, right).

File format

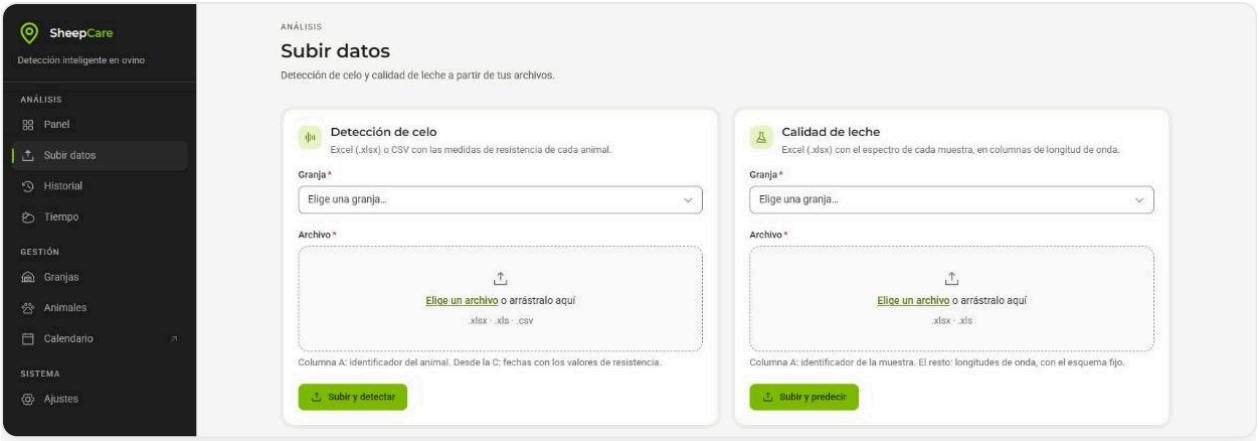
Excel (`.xlsx` , `.xls`) and CSV (`.csv`) files are accepted:

Column A	Following columns
Animal identifier (tag number)	The numeric resistance values, one per column, from left to right. The header may contain dates.

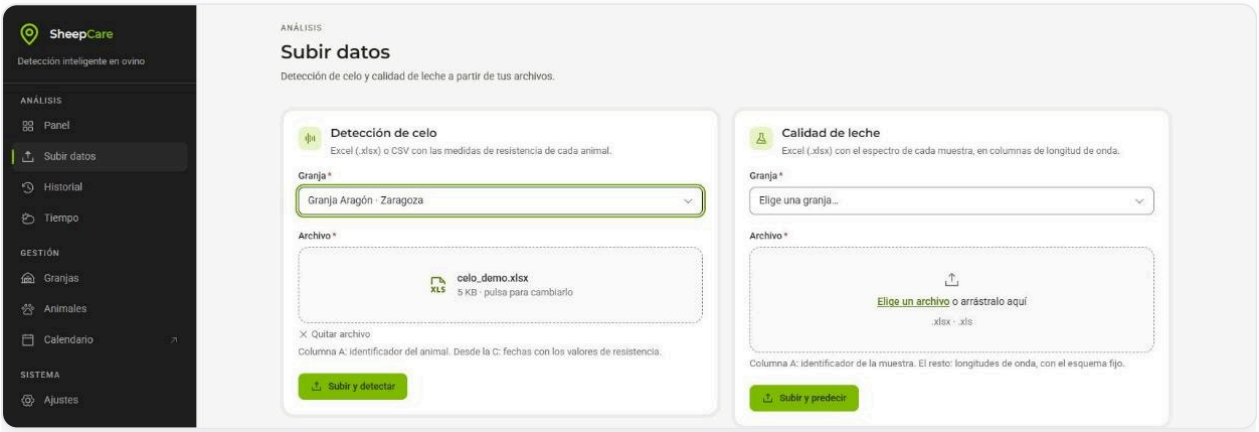
- Each animal must have **exactly 2 numeric readings**. Animals with a different number of readings are skipped.
- Non-numeric cells are ignored.
- The first row is treated as a header and skipped if cell A1 reads `ID_ANIMAL` , `ANIMAL` , `ID` , `ANIMAL ID` or `IDENTIFICACIÓN` .

Upload process

1. In the sidebar, select "Subir datos".
2. In the "Detección de celo" card, choose the **farm** to which the animals belong.
3. Click "Elige un archivo" (Choose a file) or drag the file onto the indicated area.
4. Click "Subir y detectar" (Upload and detect). The results appear on screen within a few seconds.



"Subir datos" screen with the estrus and milk quality cards.

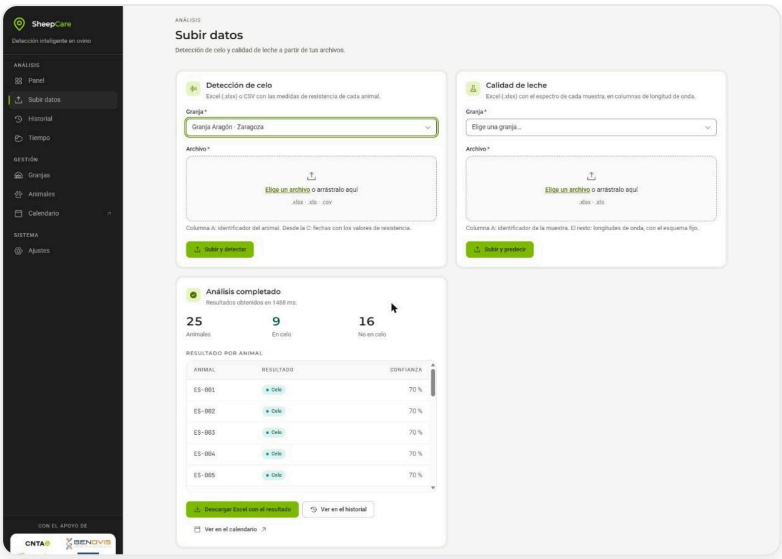


With the farm and file chosen, the card is ready to send.

Interpreting results

Result	Meaning	Label
Celo (Estrus)	The animal shows signs of being in estrus	Celo
No celo (No estrus)	The animal shows no signs of estrus	No celo

The **Confianza** (Confidence) column shows the model's certainty for each result.



Result: 9 animals in estrus and 16 without estrus out of 25, with the confidence of each one.

Available actions after detection

- **Descargar Excel con el resultado** (Download Excel with the result): downloads the original file with a *Resultado* column added at the end (*Celo* or *No celo*).
- **Ver en el historial** (View in history): opens the history with all results.
- **Ver en el calendario** (View in calendar): opens the Farm Calendar, where the detection is already recorded.

5.4 Upload Data: Milk Quality

The "Calidad de leche" card classifies the somatic cell count (SCC) of each milk sample from its near-infrared spectrum, independently of estrus detection.

File format

Unlike the estrus file, the spectral file has a fixed layout: only Excel (`.xlsx` , `.xls`) is accepted; the first column must be named `id` and the following columns must be exactly the wavelength columns the model was trained on. If the file does not match, SheepCare rejects it with a clear error instead of producing unreliable predictions.

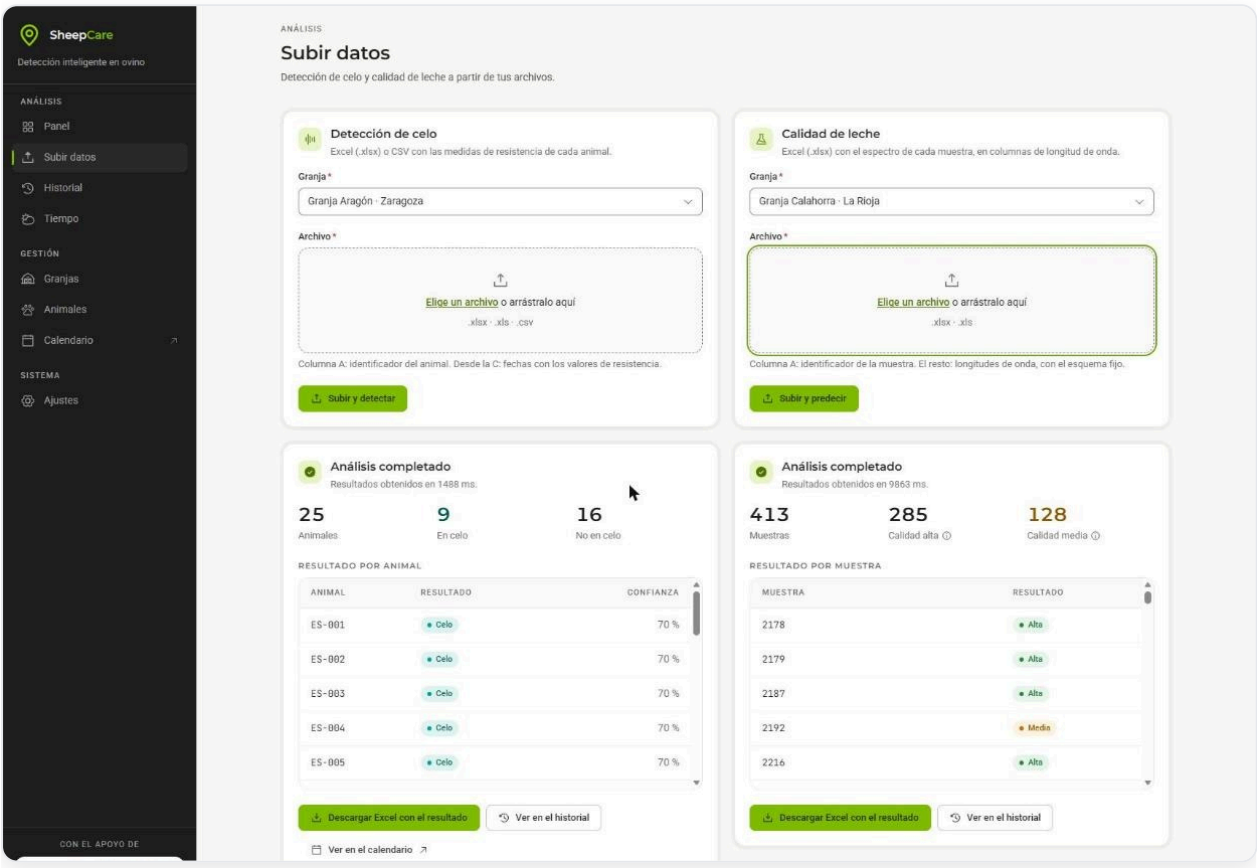
Upload process

1. In the "Calidad de leche" card, choose the farm the samples belong to.
2. Choose the spectral Excel file.
3. Click "Subir y predecir" (Upload and predict). It can take a few seconds; a file with 413 samples took about 10 seconds on the test machine.

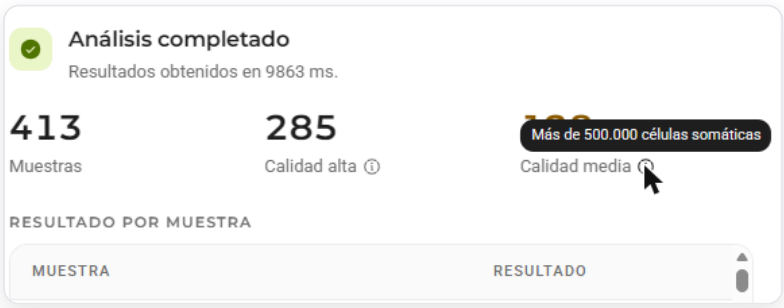
Interpreting results

Result	Meaning	Label
Calidad alta (High quality)	Fewer than 500,000 somatic cells	Alta
Calidad media (Medium quality)	More than 500,000 somatic cells: worth a closer look	Media

Next to "Calidad alta" and "Calidad media" there is an information icon that shows the threshold. Unlike estrus detection, this model **does not report a confidence percentage** for each sample.



Results of both cards: for milk, 413 samples with 285 of high quality and 128 of medium quality.



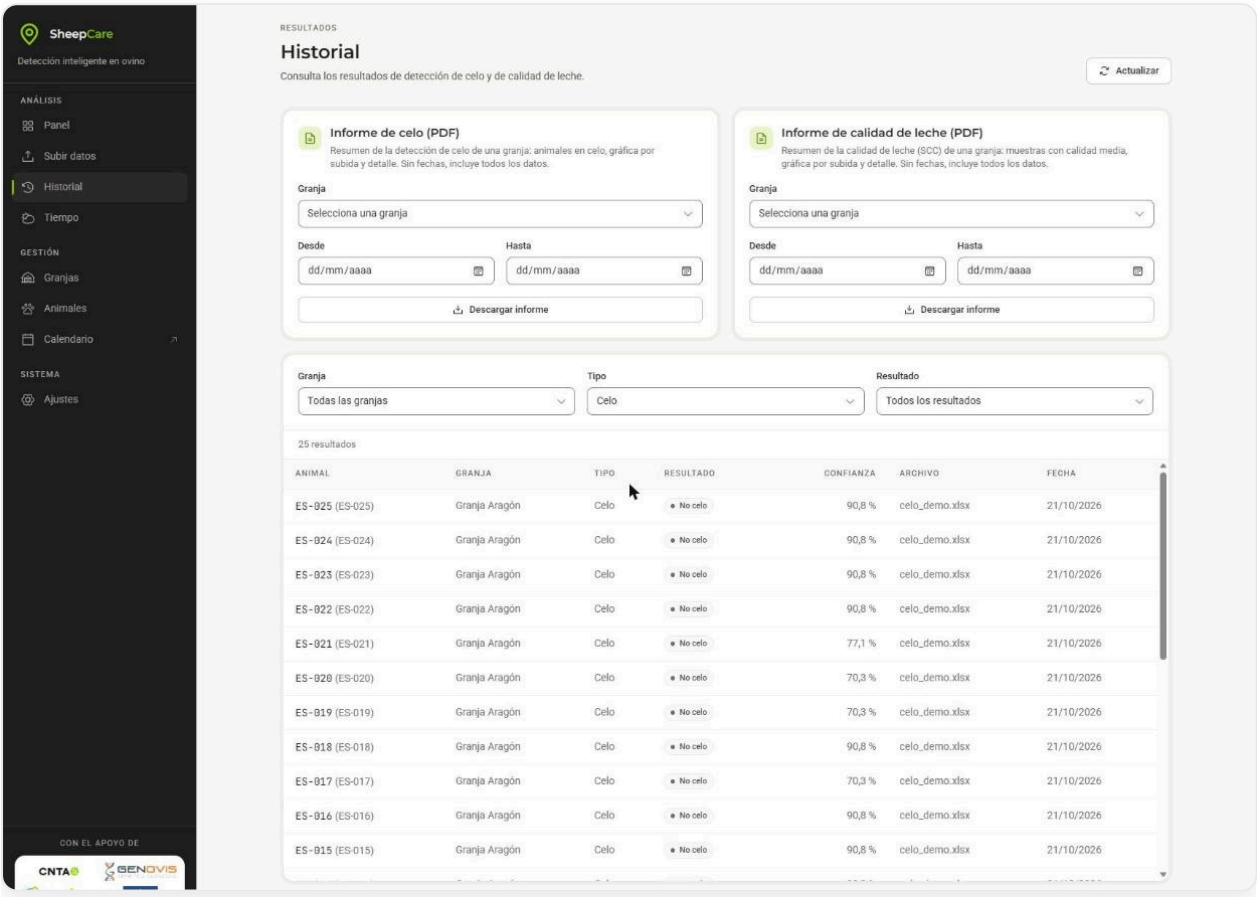
The information icon next to "Calidad media" shows the threshold.

As with estrus, you can go to the history or download the Excel file with the result. In that Excel file, the *Resultado* column contains `Lower` (high quality) or `Upper` (medium quality).

5.5 History ("Historial")

It gathers all results, estrus and milk quality, in a single list. You can filter by **farm**, by **type** (Celo / Calidad de leche) and by **result**. Above the table, a line states how many results there are.

The table includes the animal or sample, farm, type, result, confidence (estrus only), source file and date. Results are shown **200 at a time**: at the end of the list, the "Mostrar 200 más" (Show 200 more) button adds more rows.



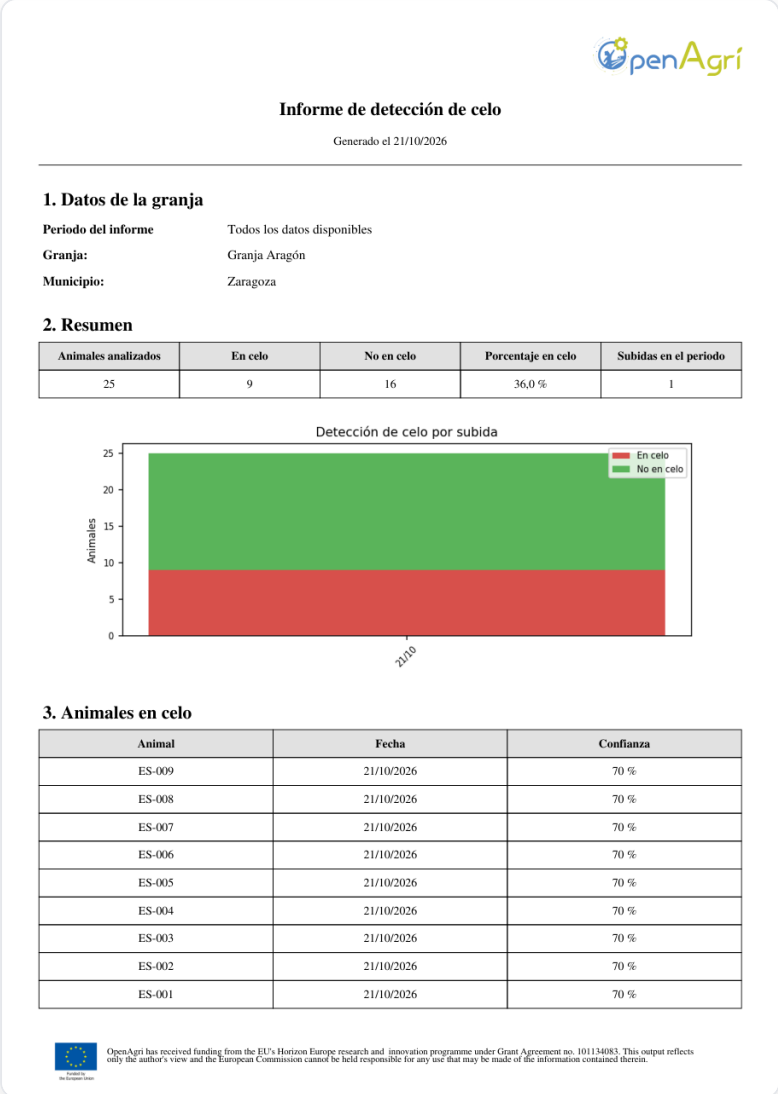
History filtered by type "Celo". At the top, the cards to generate PDF reports.

5.6 PDF Reports

At the top of the History screen there are two cards to generate per-farm reports: **Informe de celo** (Estrus report) and **Informe de calidad de leche** (Milk quality report).

1. Choose the **farm**.
2. If you wish, set the **Desde** (From) and **Hasta** (To) dates. Without dates, the report includes all of the farm's data.
3. Click "Descargar informe" (Download report). The PDF may take a few seconds to be generated.

The **estrus report** includes the farm details, a summary (animals analysed, in estrus, not in estrus, percentage and uploads), a chart per upload and the list of animals in estrus with their date and confidence. The **milk quality report** includes the farm details, the summary (samples, high quality, medium quality and percentage of medium quality), a note on the threshold, a chart per upload and the list of medium-quality samples. Both reports carry the OpenAgri logo and the European Union funding acknowledgement. The reports are written in Spanish.



OpenAgri

Informe de detección de celo

Generado el 21/10/2026

1. Datos de la granja

Periodo del informe	Todos los datos disponibles
Granja:	Granja Aragón
Municipio:	Zaragoza

2. Resumen

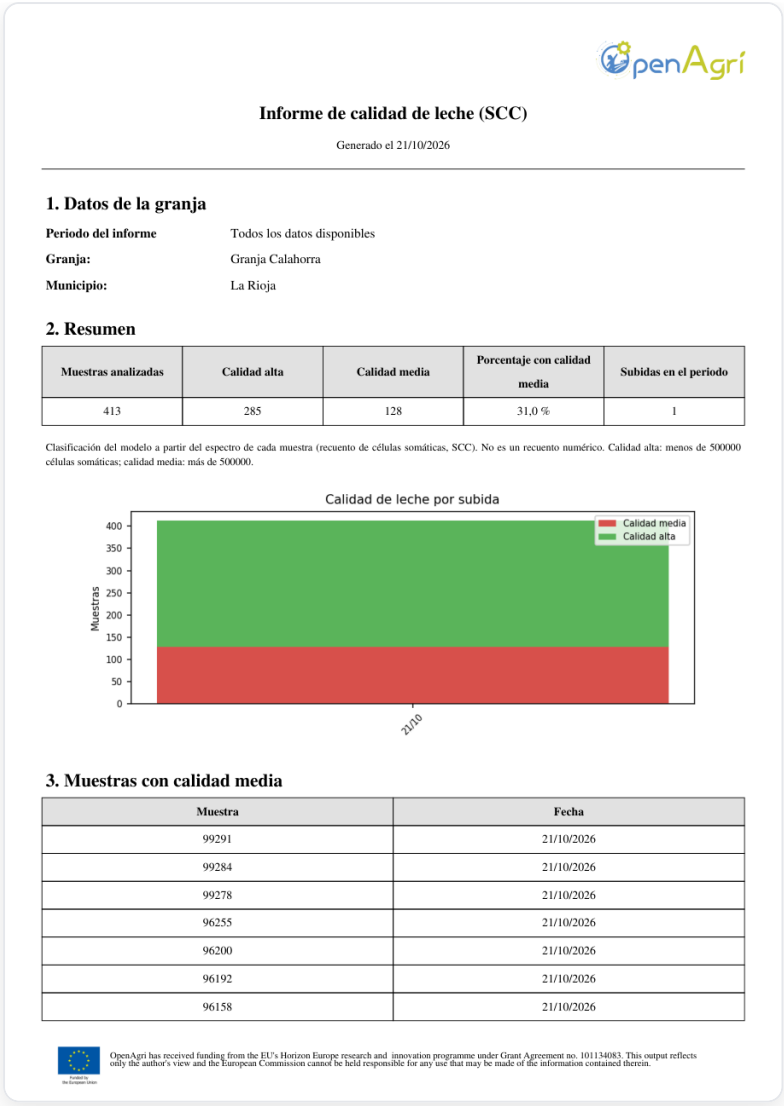
Animales analizados	En celo	No en celo	Porcentaje en celo	Subidas en el periodo
25	9	16	36,0 %	1

3. Animales en celo

Animal	Fecha	Confianza
ES-009	21/10/2026	70 %
ES-008	21/10/2026	70 %
ES-007	21/10/2026	70 %
ES-006	21/10/2026	70 %
ES-005	21/10/2026	70 %
ES-004	21/10/2026	70 %
ES-003	21/10/2026	70 %
ES-002	21/10/2026	70 %
ES-001	21/10/2026	70 %

OpenAgri has received funding from the EU's Horizon Europe research and innovation programme under Grant Agreement no. 101134083. This output reflects only the author's view and the European Commission cannot be held responsible for any use that may be made of the information contained therein.

First page of an estrus report (demonstration data).



Informe de calidad de leche (SCC)

Generado el 21/10/2026

1. Datos de la granja

Periodo del informe	Todos los datos disponibles
Granja:	Granja Calahorra
Municipio:	La Rioja

2. Resumen

Muestras analizadas	Calidad alta	Calidad media	Porcentaje con calidad media	Subidas en el periodo
413	285	128	31,0 %	1

Clasificación del modelo a partir del espectro de cada muestra (recuento de células somáticas, SCC). No es un recuento numérico. Calidad alta: menos de 500000 células somáticas; calidad media: más de 500000.

3. Muestras con calidad media

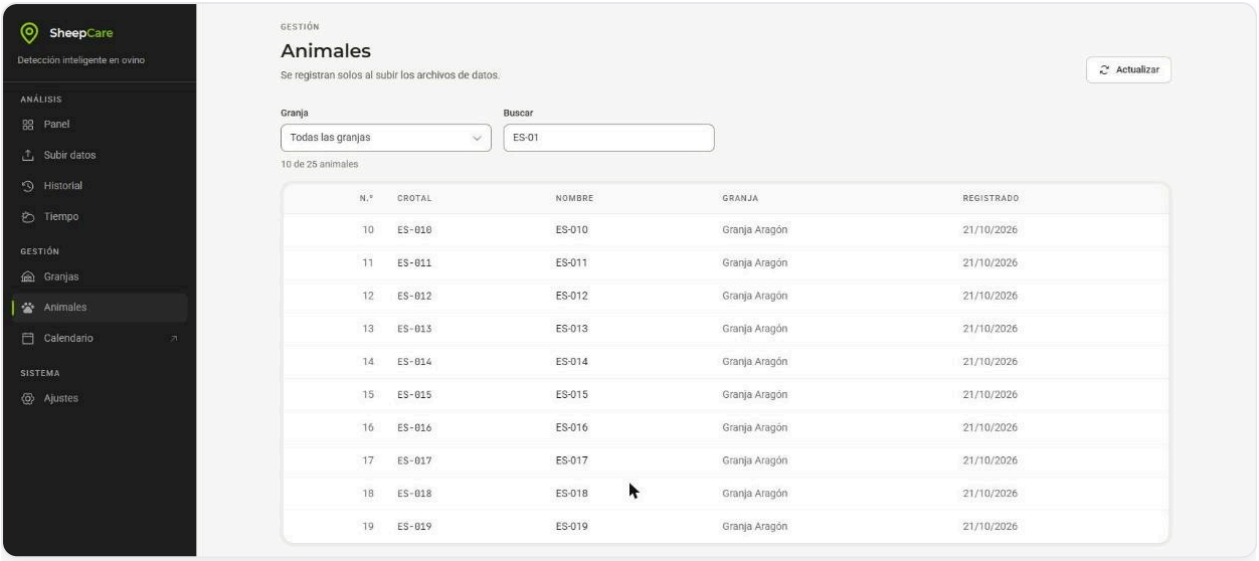
Muestra	Fecha
99291	21/10/2026
99284	21/10/2026
99278	21/10/2026
96255	21/10/2026
96200	21/10/2026
96192	21/10/2026
96158	21/10/2026

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First page of a milk quality report (demonstration data).

5.7 Animals ("Animales")

Lists all registered animals. They are not created manually: they are generated automatically when files are uploaded, using the identifier in each row. You can **filter by farm** and **search by tag or name**; above the table a line shows how many animals are visible ("10 de 25 animales"). The table header stays visible as you scroll.



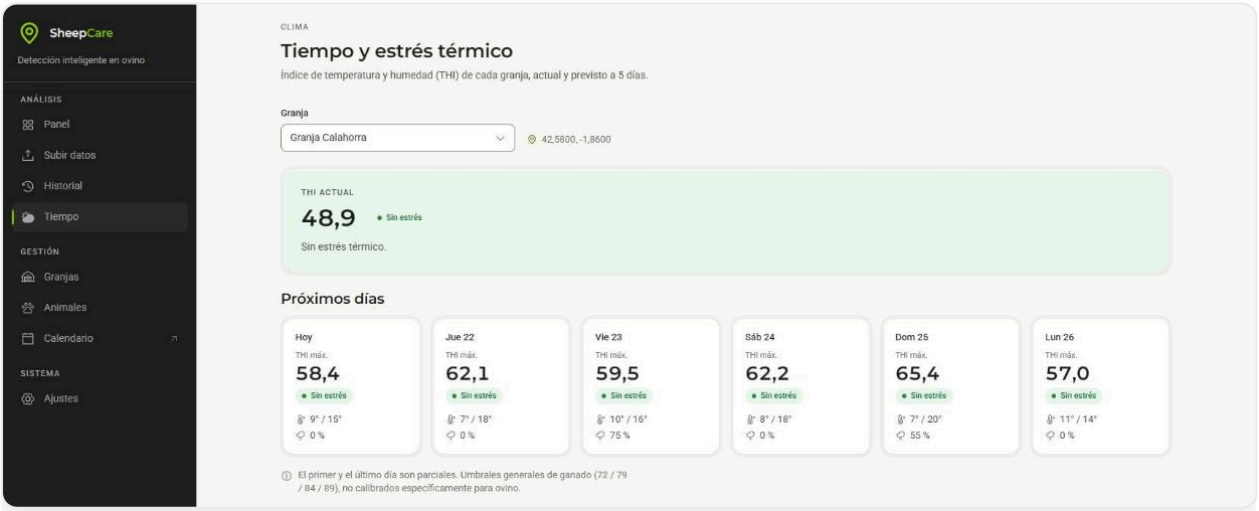
Search "ES-01": 10 of 25 animals.

5.8 Weather and Heat Stress ("Tiempo")

Shows the **temperature-humidity index (THI)** of each farm, current and forecast for 5 days, to detect heat stress in the flock. Choose the farm from the drop-down.

- **Current THI** ("THI actual"): value, level (Sin estrés / No stress, Estrés leve / mild, moderado / moderate, severo / severe or crítico / critical) and a recommendation.
- **Next days** ("Próximos días"): for each day, the maximum THI and its level, the minimum and maximum temperatures and the probability of rain. The first and last days are partial.

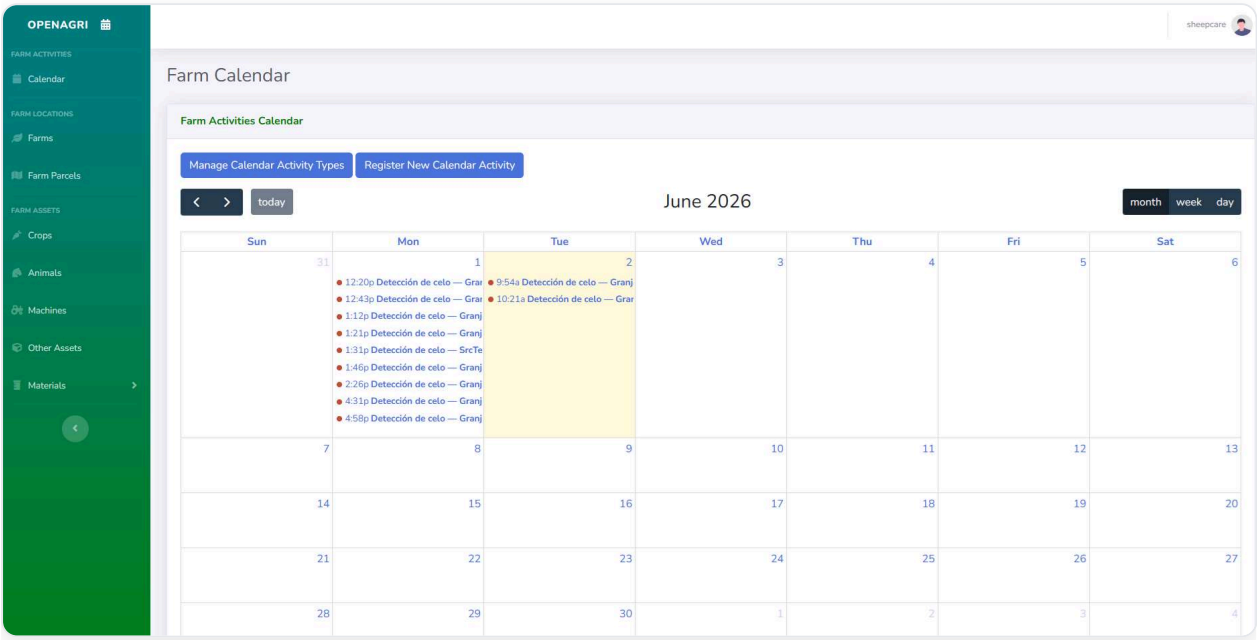
> **To use this screen you need:** the farm to have **coordinates** (section 5.2), **your own OpenWeatherMap key** saved in Settings (section 5.10) and an Internet connection. The THI thresholds are the general livestock ones (72 / 79 / 84 / 89) and are not calibrated specifically for sheep.



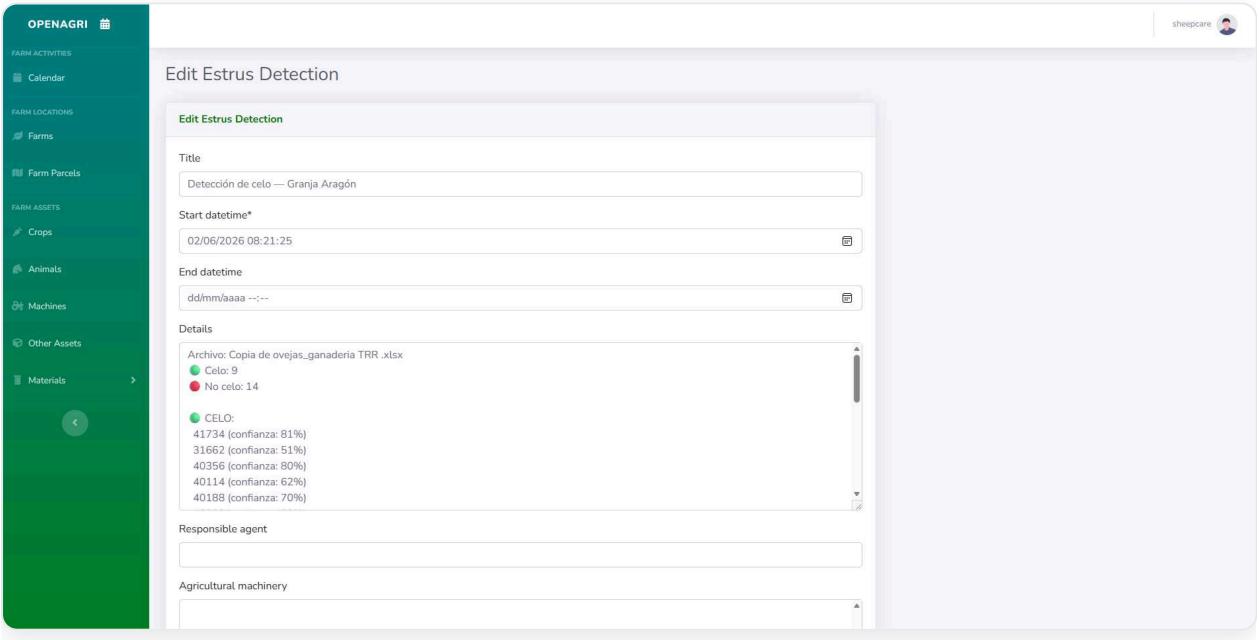
Current THI and 5-day forecast for a farm with coordinates.

5.9 Calendar and GateKeeper Login

The "Calendario" option in the sidebar opens the OpenAgri Farm Calendar module in a separate browser tab (`http://localhost:8002`). It lets you record and consult farm events in a calendar. Each time you upload a file, the results are automatically recorded in the Farm Calendar as observations: estrus results as "Estrus Detection" and milk results as "Milk Quality" (with the breakdown of high and medium quality).

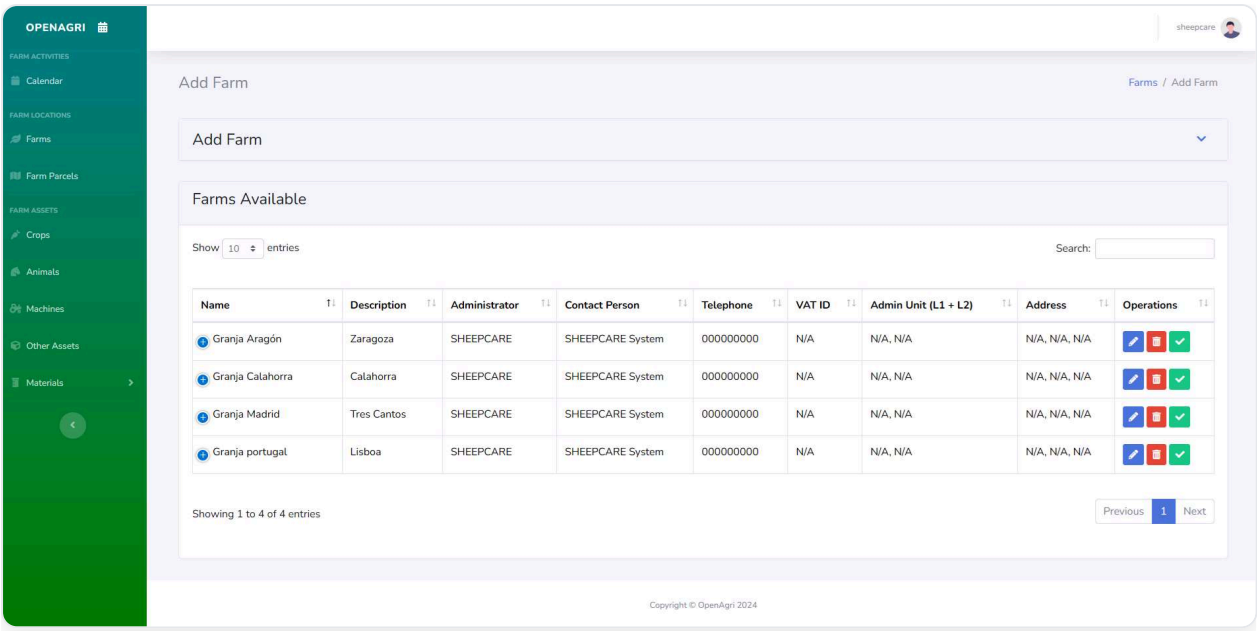


Monthly view of the Farm Calendar with the recorded detections.



Event detail: farm, date and the breakdown of results under "Details".

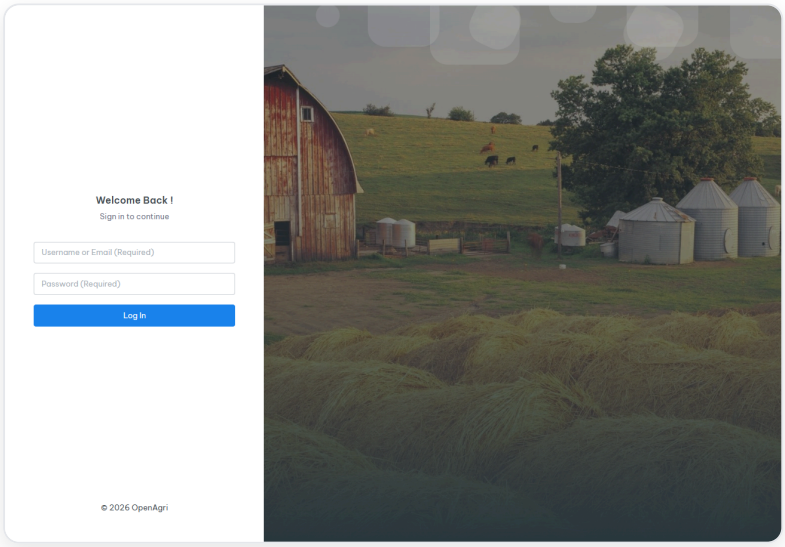
In the Farm Calendar's "Farms" section you will find the same farms you created in SheepCare, because the two modules synchronise automatically.



SheepCare farms appear in the Farm Calendar.

Logging in with GateKeeper

The Calendar has no username and password of its own: it delegates login to GateKeeper, an authentication service shared across the OpenAgri ecosystem. The first time you open the Calendar in a browser session, you are redirected to the GateKeeper sign-in page. Sign in with the administrator account created on first launch.



GateKeeper sign-in page (in English).

After a successful login, GateKeeper redirects you back to the Calendar, already authenticated. The session stays active while the browser tab remains open; if you are asked to log in again, use the same credentials.

> **Forgot your GateKeeper password?** It is stored only on this device. Ask your system administrator to check `%APPDATA%\SheepCare\gatekeeper_admin.key` on this computer, or to delete that file so that SheepCare prompts you to create a new administrator account the next time it starts.

5.10 Settings ("Ajustes")

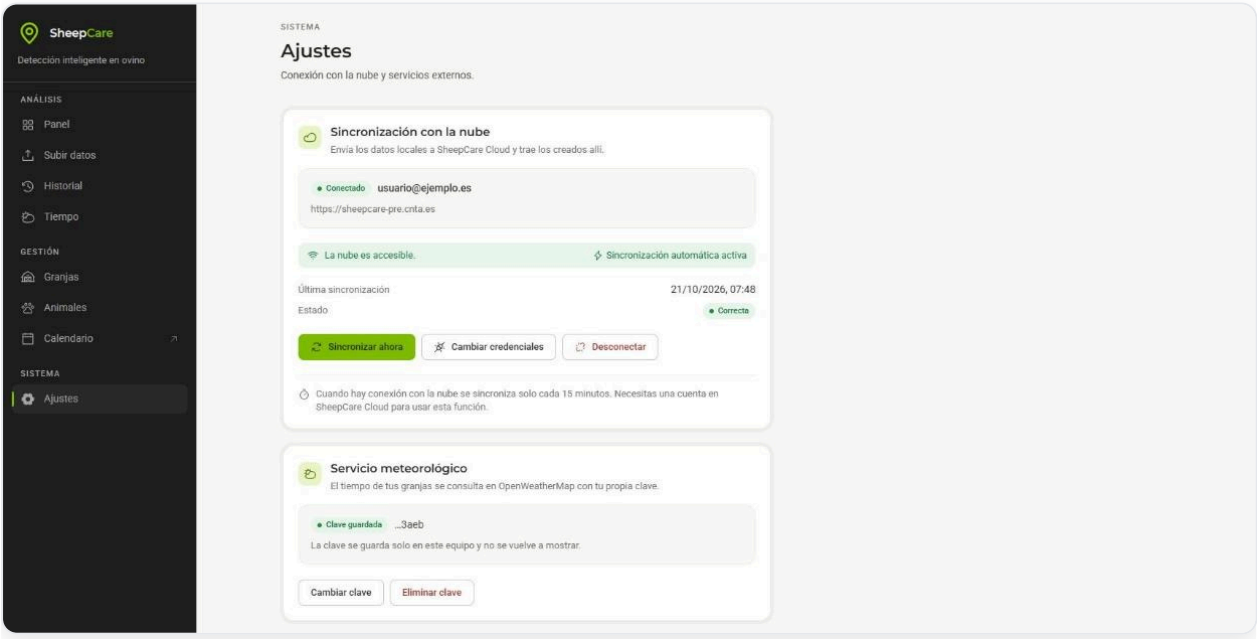
The screen has two sections.

Cloud synchronisation ("Sincronización con la nube")

It connects this device to SheepCare Cloud (section 7). Once the connection is configured, the screen shows:

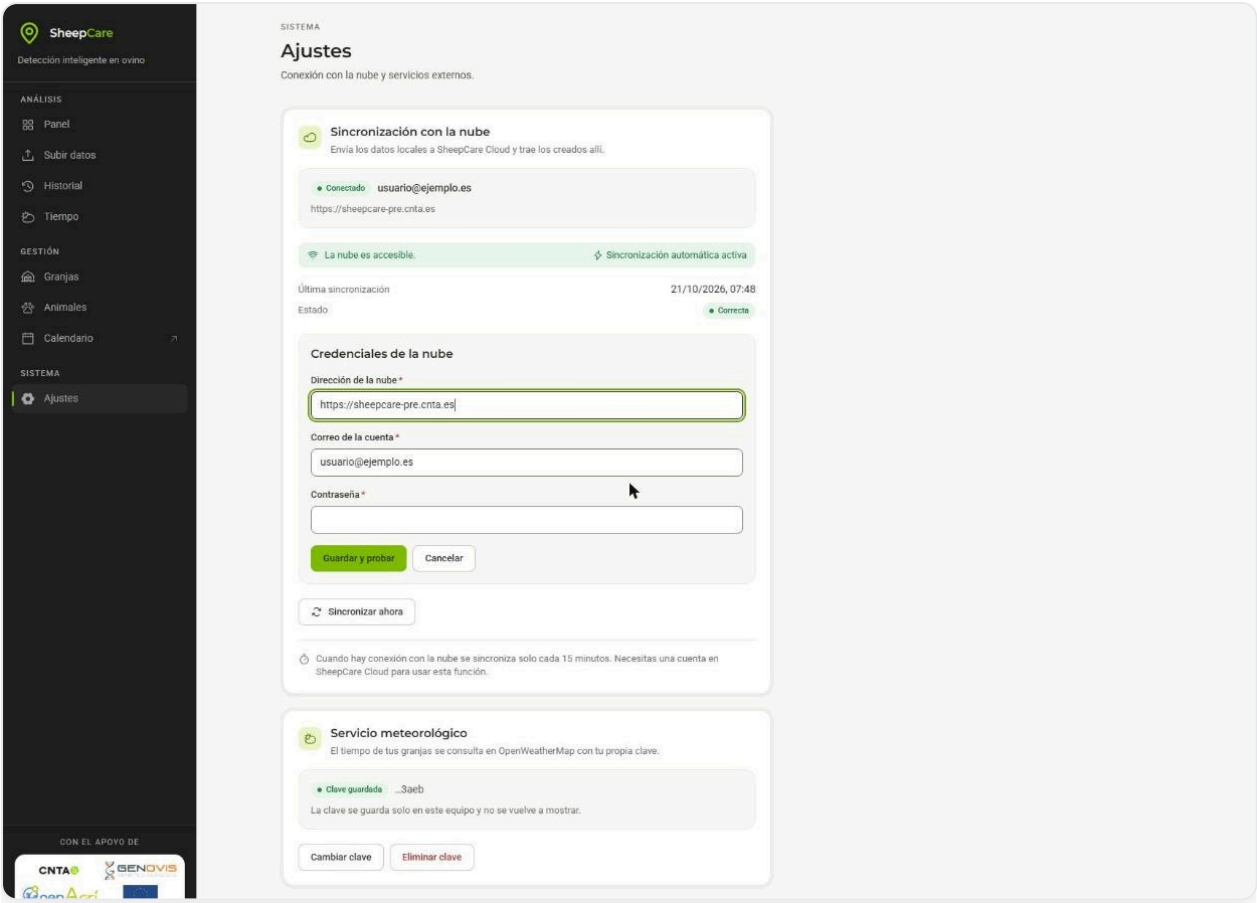
- the **Conectado** (Connected) status, the account and the server address;
- whether the cloud is reachable and whether **automatic synchronisation** is active. If there is no connection, it reads "La nube no es accesible. Solo se puede sincronizar a mano" (the cloud is not reachable, manual sync only);
- the **date of the last synchronisation** and its status (*Correcta* / successful or *Con errores* / with errors; if there was an error, a collapsible technical detail);
- a notice, if any, about data that was not sent for lack of permission (section 7).

The buttons are **Sincronizar ahora** (Sync now), **Cambiar credenciales** (Change credentials) and **Desconectar** (Disconnect: it deletes the credentials; your local data is not lost).



Settings with synchronisation connected and the weather service key saved.

To connect, click "Configurar la nube" (or "Cambiar credenciales"), enter the server address, your email and your password, and click "Guardar y probar" (Save and test). The test server address is `https://sheepcare-pre.cnta.es`; use the one CNTA gives you for your organisation.



Credentials form. "Guardar y probar" checks the access and launches a synchronisation.

Weather service ("Servicio meteorológico")

The weather for your farms is looked up in OpenWeatherMap with **your own key**. If you do not have one yet, "Sin clave" (No key) is shown: you can get a free one by registering at openweathermap.org. Click "Añadir clave" (Add key), or "Cambiar clave" (Change key) if one is already saved, paste it and click "Guardar clave" (Save key). The key is stored only on this device and is never shown again (only its last characters). "Eliminar clave" deletes it.

6. Typical Workflow

1. **Register the farm**, the first time, in "Granjas". Add its coordinates if you want to check the weather.
2. **Prepare the file**: export the Excel or CSV file (estrus) or the spectral Excel file (milk) from the measuring device.
3. **Upload and analyse** in "Subir datos": choose the card, the farm and the file.
4. **Review the results**: which animals are in estrus and with what confidence, or the quality of each milk sample.

5. **Browse the history** and filter by farm, type or result.
6. **Generate the PDF report** to archive it or share it with the responsible technician, and download the Excel file with the result if you need it.
7. **Note in the calendar** the corresponding matings or treatments, and check the farm's **weather** when needed.

7. Cloud Synchronisation

Cloud synchronisation makes the data recorded on this device available on the central SheepCare Cloud platform as well, and vice versa. It is optional and does not affect the local operation of the application. You need a SheepCare Cloud account, which is created by your organisation's administrator.

Connecting

1. Go to "Ajustes".
2. Click "Configurar la nube" and enter the server address, your email and your SheepCare Cloud account password.
3. Click "Guardar y probar". If the credentials are valid, an immediate synchronisation is launched.

When it synchronises

- **When saving credentials**: immediately.
- **On application start**: about 30 seconds after startup, if credentials are configured and the cloud is reachable.
- **Automatically every 15 minutes**, while the application is running and there is a connection.
- **Manually**: the "Sincronizar ahora" button forces a synchronisation at any time.

Farms, animals and the results for **estrus and milk quality** are synchronised, in both directions: what is created here goes up to the cloud and what is created there is downloaded. What arrives from the cloud is also recorded in the local Farm Calendar. If there is no connection, the application keeps working and synchronises as soon as it is restored.

Shared farms and permissions

Your account sees its own farms and the farms other people in the organisation share with you.

- A shared farm carries the label "Compartida contigo": you can view its data and upload files to it, but you cannot edit or delete it.

- If a farm stops being shared with you, it is **hidden on this device, not deleted**. If it is shared again, it reappears with its history.
- If you send data for a farm on which you no longer have permission, the cloud rejects it. It is **set aside**, kept on this device, not retried, and Settings tells you how many items there are.

8. Frequently Asked Questions

I uploaded an estrus file but no results appear, or there is an error

Check the format: column A with the animal identifier, numeric readings in the following columns, **exactly 2 readings per animal**, and extension `.xlsx`, `.xls` or `.csv`. Animals with a different number of readings are skipped.

The milk quality upload is rejected with a column error

The spectral file requires an exact layout: an `id` column followed by the specific wavelengths the model was trained on. Files exported from a different device, or with an edited or reordered header, are rejected. Use the export as generated by the spectrometer.

"Tiempo" says the key is missing or the farm has no coordinates

Add the OpenWeatherMap key in "Ajustes" and the farm's coordinates in "Granjas". The screen offers a direct link to each. You need an Internet connection.

The PDF report is not generated

The report is prepared by a local service that starts together with SheepCare. If "El servicio de informes no está disponible ahora mismo" (the reporting service is not available right now) appears, wait a minute and try again; if it does not resolve, restart SheepCare.

A farm has disappeared from the list

If it was a farm shared with you, it may have stopped being shared. It is hidden but not deleted: it will reappear, with its history, if it is shared again in the cloud and you synchronise.

The Calendar keeps asking me to log in / redirects in a loop

Make sure you are entering the GateKeeper administrator credentials created on first launch. If GateKeeper itself failed to start, the Calendar falls back to its own local login screen: restart SheepCare and try again.

The Farm Calendar does not load or shows an error

It may take a few extra seconds to become available after startup compared to the main application. Wait 20–30 seconds and reload the page.

Will my data be deleted if I uninstall the application?

No. Uninstalling removes the application files (`C:\Program Files (x86)\SheepCare\`), but the data (database, configuration and calendar data) remains in `%APPDATA%\SheepCare\` . To remove it completely, delete that folder manually after uninstalling.

Can I use SheepCare without an Internet connection?

Yes. Estrus detection, milk quality, history and reports work offline. Only cloud synchronisation and the weather lookup need Internet.

How do I stop the application?

Closing the browser does not stop the SheepCare services: they keep running in the background. To stop them, end the `sheepcare-backend.exe` , `sheepcare-calendar.exe` , `sheepcare-gatekeeper.exe` , `sheepcare-weather.exe` and `sheepcare-reporting.exe` processes from the Windows Task Manager, and then `mongod.exe` and `postgres.exe` ; or restart the computer. They use very few resources and in most cases it is not necessary to stop them manually.